Abstract Algebra II
Problem Set 3: Rings and Fields

Definition 0.1. An integral domain R (a ring that is commutative and has no zero divisors) is called a *Euclidean domain* if there is at least one function $f := R \setminus \{0\} \to \mathbb{N}$ such that, for any $a \in R$ and any non zero $b \in R$, there are elements $p, q \in R$ such that

$$a = p \cdot b + q$$

where either $q = 0$ or $f(q) < f(b)$.

Example 0.2.
1. $\mathbb{Z}$ is a Euclidean domain, where one possible f is $f(n) = |n|$.
2. $\mathbb{Z}[\sqrt{-5}]$ is *not* a Euclidean domain.

Question 1. Prove that any polynomial ring $F[x]$ over a field F is a Euclidean domain.
(Hint: for a polynomial $p(x) \in F[x]$ let $f(p) = \deg(p)$.)

Question 2. Use Eisentstein's criterion to show that cyclotomic polynomials

$$\frac{x^p - 1}{x - 1} = x^{p-1} + \ldots + x + 1$$

are irreducible in $\mathbb{Q}[x]$.
(Hint: Use the substitution $x = y + 1$.)

Question 3. Show $\big(\mathbb{Z}[\sqrt{2}]\big)[\sqrt{3}] = \mathbb{Z}[\sqrt{2}, \sqrt{3}]$.

Question 4. Prove that in a principal ideal domain every non-zero non-unit element can be written as a product of prime elements.